

MULTIMEDIA

What is Multimedia ??

Multimedia is the field concerned with the computer-controlled integration of text, graphics, drawings, still and moving images (Video), animation, audio, and any other media where every type of information can be represented, stored, transmitted and processed digitally.

What is Multimedia ??

Multimedia includes a combination of text, audio, still images, animation, video, and **interactivity** content forms.

A ***Multimedia Application*** is an Application which uses a collection of multiple media sources e.g. text, graphics, images, sound/audio, animation and/or video.

What is Multimedia ??

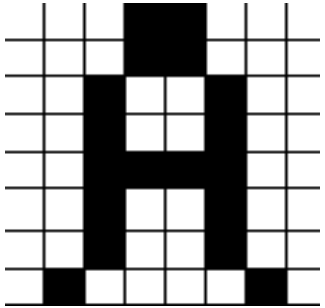


Text

- When PC's were in their start running under MS-DOS they only displayed text in one size and one color.

Images

- Turning on or off monitor pixels in graphics mode can create an alpha numeric character. Obviously the same process was used to create a picture in the earlier computers.





Images

- All computer colors are created by mixing RED GREEN and BLUE

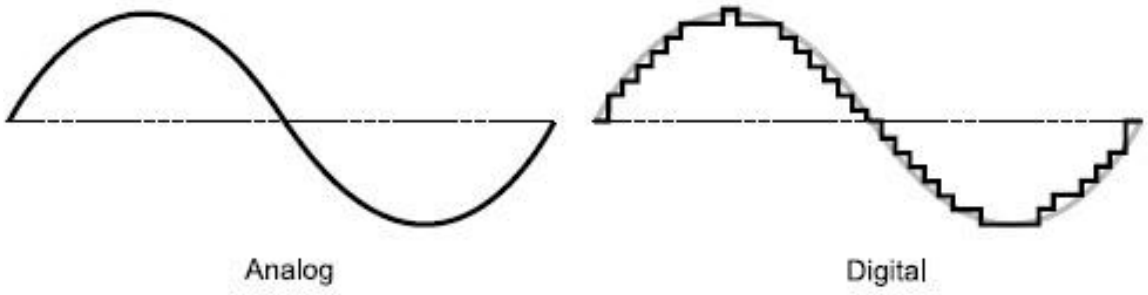


Audio

- Refers to the reproduction and transmission of sound stored in a digital format. This includes CDs as well as any sound files stored on a computer.

Audio

- Digital sound can be referred as the sound that is Recorded and stored as a series of numerical values rather than fluctuations in amplitude.



Animation

- Animation is a sequential series of still images that create an illusion of motion.



Animation

- Note the difference between animation and video.
- Whereas video takes continuous motion and breaks it up into discrete frames,
- animation starts with independent pictures and puts them together to form the illusion of continuous motion.

Animation

- 2D animation figures are created and/or edited on the computer using 2D bitmap graphics or created and edited using 2D vector graphics. 3D animation are digitally modeled and manipulated by an animator.

Video

- Unlike an animation that we can create from drawings or images a video is created by a photographic process and converted or ported to a computer in sets of Frames where each frame has data stored in every pixel.

Video

- We measure the rate at which frames are displayed in **Frames Per Second (FPS)**.

- Digital video can be copied with no degradation in quality.

Video



Video




gifbin.com

Multimedia categories

Multimedia may be broadly divided into

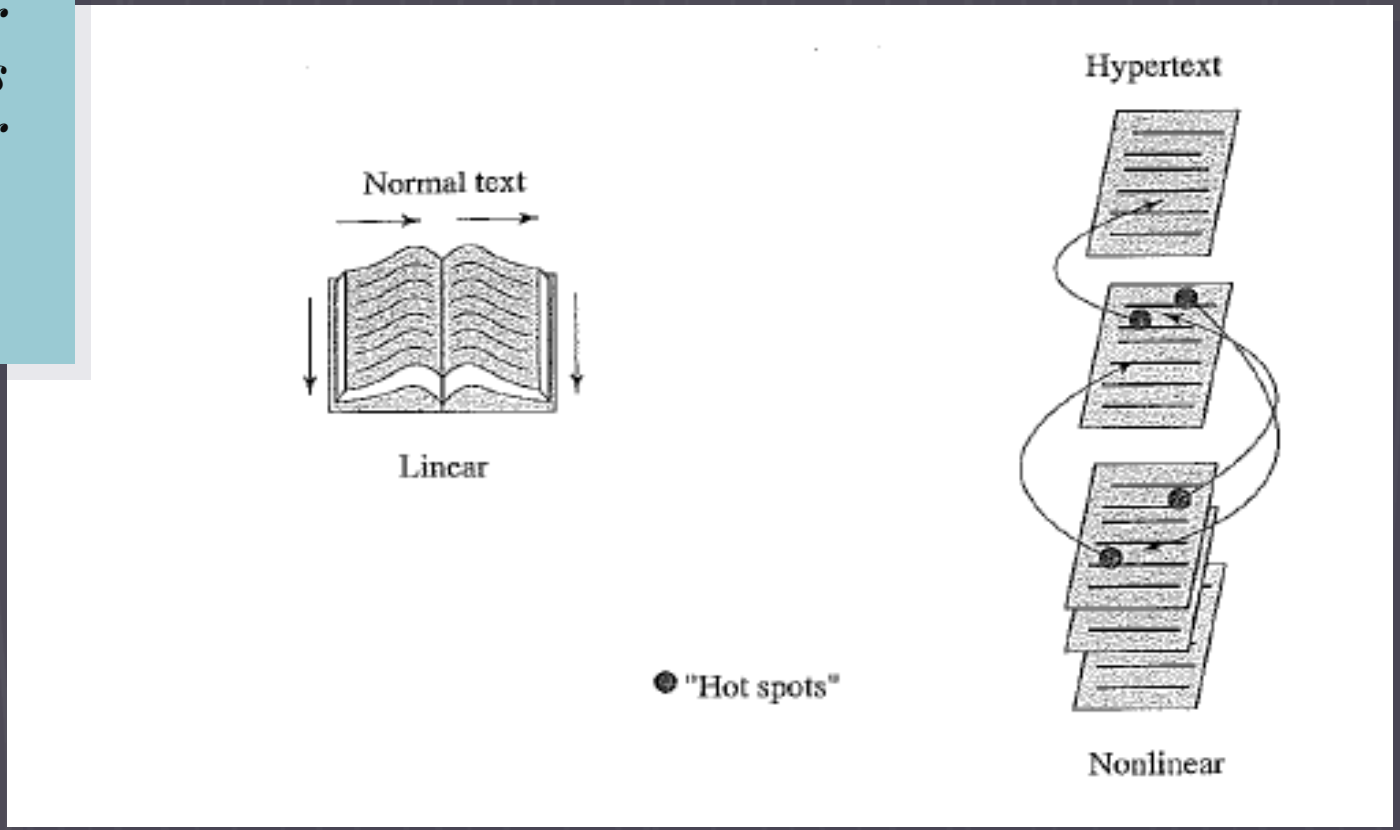
- **Linear** and
- **Non-linear** categories.

Multimedia categories

- Linear active content progresses without any navigational control for the viewer such as a cinema presentation.

- Non-linear content offers user interactivity to control progress as used with a computer game or used in self-paced computer based training.

Linear vs Non-Linear



Hypertext

- *Hypertext* is a text which contains links to other texts.
- The term was invented by Ted Nelson around 1965.



Hypermedia

- HyperMedia is not constrained to be text-based. It can include other media, e.g., graphics, images, and especially the continuous media - sound and video.
- Apparently, Ted Nelson was also the first to use this term.
- The World Wide Web (WWW) is the best example of hypermedia applications.

Multimedia Systems

- A Multimedia System is a system capable of processing multimedia data and applications.

- A Multimedia System is characterized by the processing, storage, generation, manipulation and rendition of Multimedia information.

Components of a Multimedia System

- Components (Hardware and Software) required for a multimedia system:
- Capture devices — Video Camera, Video Recorder, AudioMicrophone, Keyboards, mice, graphics tablets, 3D input devices, tactile sensors, VR devices. Digitising Hardware

Components of a Multimedia System



Video Camera



Graphics Tablet

Components of a Multimedia System

- Storage Devices — Hard disks, CD-ROMs, DVD-ROM, etc
- Communication Networks — Local Networks, Intranets, Internet, Multimedia or other special high speed networks.

Applications of Multimedia

- World Wide Web
- Video conferencing
- Video-on-demand
- Interactive TV
- Digital Transmission
- Groupware
- Webcasting
- Home shopping
- Games
- Virtual reality
- Digital video editing and production systems

World Wide Web

- Multimedia incorporates different media: sound, images, video.
- The world wide web is a global hypermedia system.
- Animation and video can show information that is difficult to convey statically.

World Wide Web

- Applications of hypermedia include online help, education and e-commerce
- Dynamic web content can be used for simple online demonstration or for complete web-based business applications.

Video Conferencing

- It is the integration of video, audio and peripherals to enable two or more people to communicate simultaneously over some type of telecommunications lines.
- In other words, you are transmitting synchronized images and verbal communications between two or more locations as if them being in the same room.

Streaming media

- Video on Demand
- Interactive TV
- Digital Transmission - Can also be referred as Streaming Media
- Streaming media is multimedia that is constantly received by and presented to an end-user while being delivered by a streaming provider

Multimedia – used in

- Multimedia finds its application in various areas including, but not limited to,

- Advertisements,
- Art,
- Education,
- Entertainment,
- Engineering,
- Medicine
- Mathematics
- Business
- Scientific research

Multimedia Software Tools

Categories of software tools

- Music sequencing and Notation
- Digital Audio
- Graphics and Image Editing
- Video Editing
- Animation
- Multimedia Authoring

Music sequencing and Notation

A music sequencer (or just sequencer) is a computer program or electronic device for recording, editing and playing back music, in the form of digital audio and/or MIDI data.

Music sequencing and Notation

- Adobe Audition - Adobe Systems
- DirectMusic Producer – Microsoft
- Deluxe Music Construction Set - Electronic Arts
- GarageBand - Apple
- Sonar – Cakewalk
- MIDI Converter Studio - ManiacTools

Digital Audio

- A digital audio editor is a computer application for audio editing, i.e. manipulating digital audio. Digital audio editors are the main software component of a digital audio workstation.
- Audition - Adobe Systems
- Creative Wavestudio - Creative Technology
- Sound Forge - Sony

Graphics and Image Editing

In computer graphics, graphics software or image editing software is a program or collection of programs that enable a person to manipulate visual images on a computer.

- Computer graphics can be classified into two distinct categories: Raster graphics and Vector graphics.

Raster Graphics

- A raster graphics editor is a computer program that allows users to paint and edit pictures interactively on the computer screen and save them in one of many popular “bitmap” or “raster” formats such as JPEG, PNG, GIF and TIFF.
- Usually an image viewer is preferred over a raster graphics editor for viewing images.

Vector Graphics Editor

- A vector graphics editor is a computer program that allows users to compose and edit vector graphics images interactively on a computer and save them in one of many popular vector graphics formats, such as EPS, PDF, WMF, SVG, or VML.

Bitmap editors

- Bitmap editors are more suitable for retouching, photo processing, photorealistic illustrations, collage, and illustrations drawn by hand with a pen tablet.

Bitmap editors

- Recent versions of bitmap editors such as GIMP and Photoshop support vector tools (e.g. editable paths), and vector editors such as CorelDRAW, Adobe Illustrator, Xara Xtreme, Adobe Fireworks, Inkscape or SK1 are adopting raster effects that were once limited to bitmap editors (e.g. blurring).

Video Editing

- **Video editing software**, also known as Non Linear Editing (NLE), is application software which handles the editing of video sequences on a computer. NLE systems replace traditional flatbed celluloid film editing tools and analogue video tape-to-tape machines.

Video Editing - Software

- These software applications allow non-linear editing of videos:

- Adobe Premiere (Windows)
- Final Cut (Mac OS X) – Apple
- VideoStudio (Windows) - Ulead Studio
- After Effects

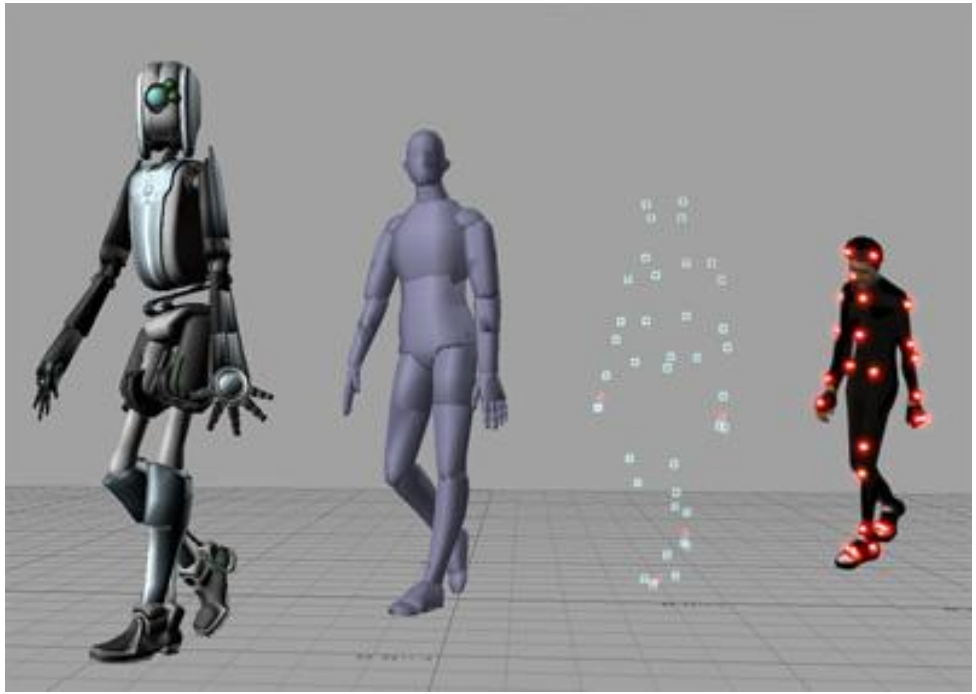
Animation

- Computer animation is the process used for generating animated images by using computer graphics. The more general term computer generated imagery encompasses both static scenes and dynamic images, while computer animation only refers to moving images produced by exploiting the persistence of vision to make a series of images look animated.

Animation

- Given that images last for about one twenty-fifth of a second on the retina fast image replacement creates the illusion of movement.

Animation





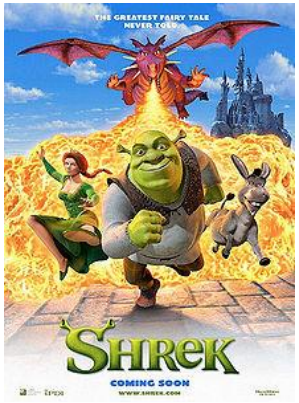
Animation

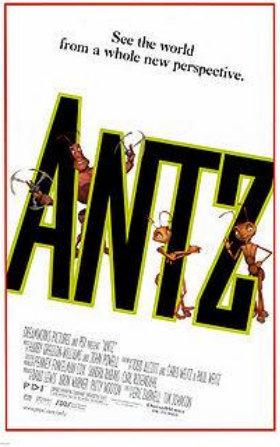


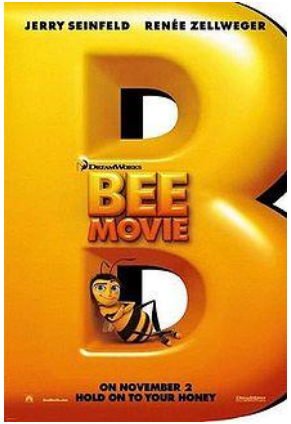
Animation of a Skeleton

Animation Film – Examples

- *Antz*
- *Shrek*
- *Bee Movie*
- *Zootopia*
- *Minions*
- *Rio*







List of Animation Softwares

- Adobe Animator
- Autodesk Maya
- Cinema 4d
- 3ds max
- Blender

Identifying various Multimedia Aspects

- In this section, we will try to understand the input methods, how to capture and store each Media format and also whether compression is required or not .
- Input
- Storage
- Compression
- Size

Text



- **Input-** Note that text and graphics (and some images) are mainly generated directly by computer/device (e.g. drawing/painting programs) and do not require digitizing:
- They are generated directly in some (usually binary) format.
- Printed text and some handwritten text can be scanned via Optical Character Recognition(OCR)
- Handwritten text could also be digitised by electronic pen sensing

Text

- Printed imagery/graphics can be flatbed scanned directly to image formats.
- **Source:**
  - Keyboard, speech input, optical character recognition, data stored on disk.
- **Storage**:
  - 1 byte per character (text or format character), e.g. ASCII.
  - For other forms of data (e.g. Spreadsheet files). May store as text (with formatting) or may use binary encoding.

Text



- Formatted Text: Raw text or formatted text e.g HTML, Rich Text Format (RTF),Word or a program language source (C, Java, etc).
- Data Not temporal—BUT may have natural implied sequence e.g. HTML format sequence, Sequence of C program statements.

Identifying various Multimedia Aspects

- Size: Not significant with respect to other Multimedia data formats.

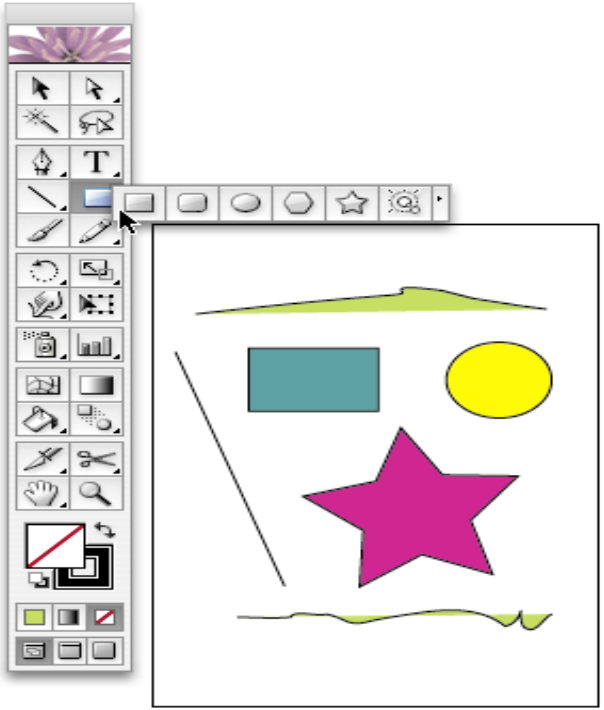
- Compression : convenient to bundle files for archiving and transmission of larger files. E.g. Zip.

Graphics

- **Format**:
  - constructed by the composition of primitive objects
  - such as lines, polygons, circles, curves and arcs.

- **Input**:
  - Graphics are usually generated by a graphics editor
  - program (e.g. Illustrator, Freehand) or automatically by a program (e.g. Postscript).

Graphics



- Graphics input devices:

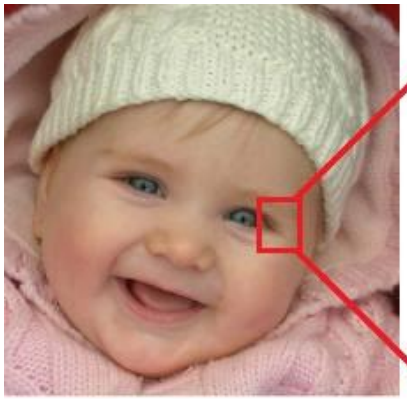
- keyboard (for text and cursor control), mouse, trackball or graphics tablet.

- Graphics are usually selectable and editable or revisable (unlike images).

- **Compression :**
- Do not take up a very high storage overhead.

Images

- Still pictures which (uncompressed) are represented as a bitmap (a grid of pixels).



75	55	41	96	97	38	33
39	75	58	48	68	6	62
91	88	78	58	94	5	51
79	93	62	48	73	88	17
50	22	76	38	2	46	68
5	21	35	52	95	79	19
76	44	51	15	98	71	33
77	71	51	20	75	9	91
4	89	54	62	67	9	65
88	18	17	46	7	16	98
56	31	44	50	29	74	66

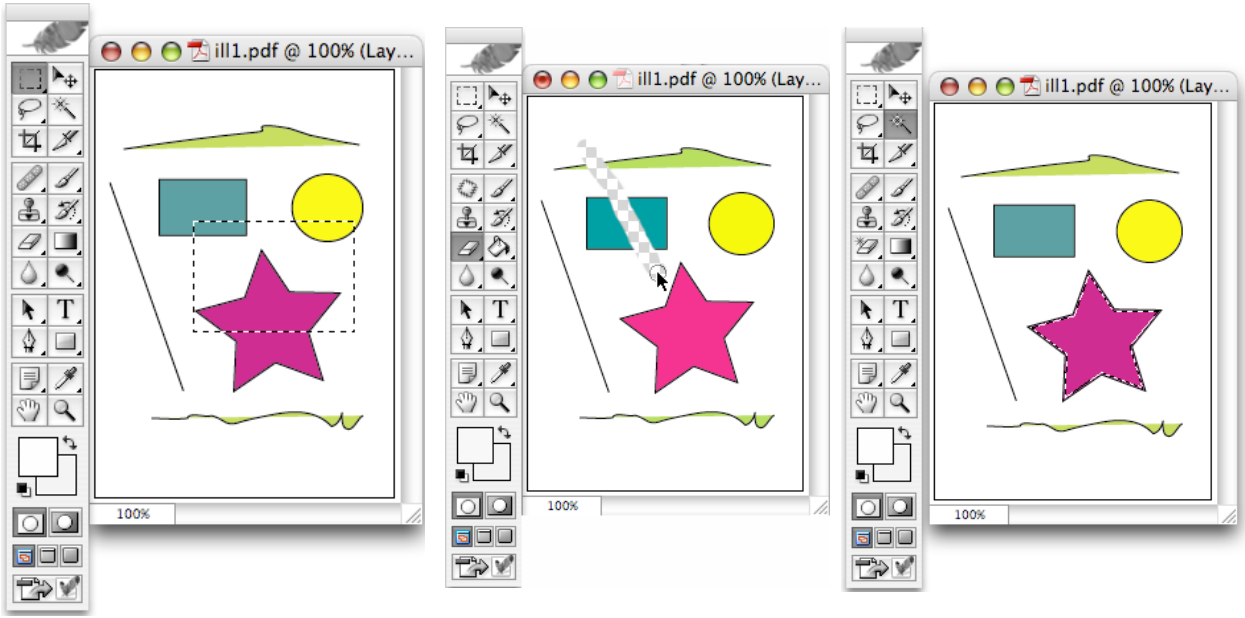
Images

- Input: Scanned for photographs or pictures using a digital
- scanner or from a digital camera.
- May also be generated by programs similar to graphics or animation programs.

Images

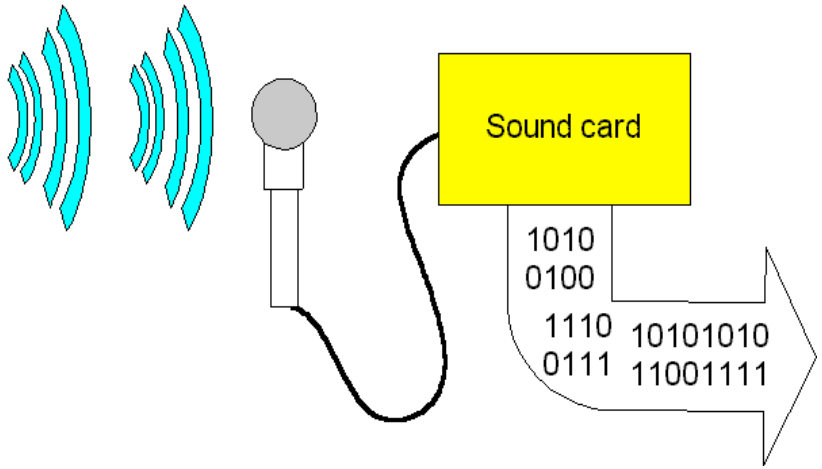
- Compression:
- Compression is commonly applied.
- Can usually only edit individual or groups of pixels in an
- image editing application, e.g. photoshop.

Images



Audio

- Audio signals are continuous analog signals.
- Input: Microphones and then digitised and stored



Audio

- Storage :
- CD Quality Audio requires 16-bit sampling at 44.1 KHz:
- Even higher audiophile rates (e.g. 24-bit, 96 KHz)

Audio

- 1 Minute of Mono CD quality (uncompressed) audio = 5 Mb.
- Stereo CD quality (uncompressed) audio = 10 Mb.
- Compression :
- Usually compressed (E.g. MP3, AAC, Flac)

Video

- **Input**:
- Analog Video is usually captured by a video camera and then digitized, although digital video cameras now essentially perform both tasks.
- There are a variety of video (analog and digital) formats
- Raw video can be regarded as being a series of single images.
- There are typically 25, 30 or 50 frames per second.

Compression

- Data Compression is the process of encoding information using fewer bits than the original representation would use.
- Compression is useful because it helps reduce the consumption of expensive resources, such as hard disk space or transmission bandwidth.
- On the downside, compressed data must be decompressed to be used, and this extra processing may be detrimental to some applications.

Compression

- There are two types of File compression algorithms:
- Lossless
- Lossy.

Compression

Feature	Lossy Compression	Lossless Compression
Data lost?	Yes	No
File size	Smaller	Larger
Quality	Slightly reduced	Original quality
Can restore original?	No	Yes
Examples	JPEG, MP3, MP4	PNG, FLAC, ZIP
Best for	Media (streaming, web)	Archiving, editing

Lossless compression

- Algorithms reduce file size without losing image quality, though they are not compressed into as small a file as a lossy compression file. When image quality is valued above file size, lossless algorithms are typically chosen.

Lossy compression

- Algorithms take advantage of the inherent limitations of the human eye and discard invisible information.
- Most lossy compression algorithms allow for variable quality levels (compression) and as these levels are increased, file size is reduced. At the highest compression levels, image deterioration becomes noticeable as "compression artifacting".

Audio compression

- Audio compression is a form of data compression designed to reduce the transmission bandwidth requirement of digital audio streams and the storage size of audio files.
- Audio compression algorithms are implemented in computer software as audio codecs.

Audio compression

- Consequently, specifically optimized audio lossless and lossy algorithms have been created. Lossy algorithms provide greater compression rates and are used in mainstream consumer audio devices.

Video compression

- Video compression refers to reducing the quantity of data used to represent digital video images, and is a combination of spatial image compression and temporal motion compensation.
- Most video compression is lossy — it operates on the premise that much of the data present before compression is not necessary for achieving good perceptual quality.

Multimedia File Formats

Image File Formats

- Image file formats are standardized means of organizing and storing digital images.
- Image files are composed of either pixel or vector (geometric) data that are rasterized to pixels when displayed (with few exceptions) in a vector graphic display.

Image file formats.

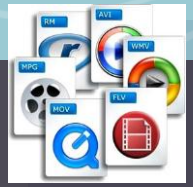
- JPEG/JFIF
- JPEG 2000
- Exif
- TIFF
- RAW
- PNG
- GIF
- BMP
- PPM, PGM, PBM, PNM
- WEBP

Audio File Formats

- An audio file format is a file format for storing audio data on a computer system.
- It can be a raw bitstream, but it is usually a container format or an audio data format with defined storage layer.
- The general approach towards storing digital audio is to sample the audio voltage which, on playback, would correspond to a certain level of signal in an individual channel with a certain resolution—the

Audio File Formats

- Waveform Audio File Format
- MP3
- Au file format
- Windows Media Audio (WMA)
- RealAudio
- The Adaptive Multi-Rate (AMR or AMR-NB)
- Advanced Audio Coding (AAC)
- MPEG-4 Part 14 or MP4 file format
- Red Book is the standard for audio CDs



Video File Formats

The Video Files category includes a wide range of video formats, which use different codecs to encode and compress video data.

Video File Formats

- The AVI Format
- The Windows Media Format
- Advances Streaming Format (.asf)
- Data File
- The MPEG Format
  - MPEG-1
  - MPEG-2
  - MPEG-3
  - MPEG-4

Video File Formats

- The Quick Time Format
- The Real Video Format
- The Shockwave Format
- Flash Video Format (.flv)
- The Matroska Multimedia Container (MKV)
- 3GPP Multimedia File